1 Minimizing complex functions with Wirtinger derivatives

1.1 Definition

Define for $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$

$$\frac{\partial}{\partial z} \equiv \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big) \tag{$*$}$$

for functions of one complex variable, and

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\Big) \qquad \& \qquad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\Big(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\Big) \tag{1}$$

for functions of multiple complex variables.

1.2 Agreement with the complex derivative

If f is complex differentiable at a point (i.e. the limit $f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$ exists), then $\frac{\partial f}{\partial z}$ agrees with the complex derivative $\frac{df}{dz}$.

Proof. For $f(z) = u(z) + iv(z)$ complex differentiable,

$$\begin{aligned} \frac{\partial f}{\partial z} &\overset{(*)}{=} \frac{1}{2}\Big(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} - i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\Big) \\ &\overset{\text{CR}}{=} \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} \overset{\text{Def.}}{=} \frac{df}{dz}, \end{aligned} \tag{2}$$

where in the second step the Cauchy–Riemann (CR) equations $\partial_x u = \partial_y v$ and $\partial_x v = -\partial_y u$ were used. □

1.3 The $\bar{z}$-derivative and Cauchy–Riemann

Furthermore,

$$\begin{aligned} 0 \overset{!}{=} \frac{\partial f}{\partial \bar{z}} &= \frac{1}{2}\Big(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\Big) = \frac{1}{2}\Big(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} + i\frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}\Big) \\ &\iff \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \wedge \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}, \end{aligned} \tag{3}$$

so that $0 = \dfrac{\partial f}{\partial \bar{z}}$ is equivalent to the Cauchy–Riemann equations, i.e. the necessary & sufficient condition for f to be complex differentiable.

1.4 Minimizing a real-valued function

Minimize a real-valued function $f\colon \mathbb{C} \to \mathbb{R}$, $f(z) = g(x, y)$, via the condition

$$\nabla g(x, y) = \begin{pmatrix} \partial_x g \\ \partial_y g \end{pmatrix} \stackrel{!}{=} 0. \tag{4}$$

(Minimizing a complex-valued f does not make sense because $\mathbb{C}$ has no order.) Then

$$\partial_x g \stackrel{!}{=} 0 \stackrel{!}{=} \partial_y g \quad \Longleftrightarrow \quad \frac{\partial f}{\partial \bar{z}} \stackrel{!}{=} 0, \tag{5}$$

so one can minimize a real-valued function by imposing $\boxed{\dfrac{\partial f}{\partial \bar{z}} = 0}$.